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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

EPH613 - APPLIED ELECTRONICS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. Distinguish FET with MOSFET.
2. What is holding current in a SCR?
3. Give any four characteristics of OP.AMP.
4. Define Common Mode Rejection Ratio.
5. What is the principle of logarithmic amplifier?
6. Distinguish between Bistable multivibrator and Schmitt trigger.
7. Name the pins of IC 555 Timer.
8. Define lock-in range of a Phase locked loops.
9. What is an accuracy of a D/A converter?
10. Differentiate between Analog and Digital signals.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Explain the function of a UJT relaxation oscillator with waveforms.

(or)

b) Explain the construction and characteristic curve of a SCR.

12. a) Draw the circuit diagram of an inverting amplifier using OP.AMP and derive the expression for its voltage gain.

(or)

b) Explain with circuit, the function of a differentiator usig OP.AMP.

13. a) Draw the circuit diagram of logarithmic divider and discuss its operation.

(or)

b) Explain with a circuit, the working of monostable multivibrator using OP.AMP.

14. a) Explain how a Astable multivibrator is constructed using IC 555 Timer.

(or)

b) Discuss the function of a voltage controlled oscillator with its circuit.

15. a) Explain with necessary theory, the working of 4-bit binary weighted D/A converter.

(or)

b) Explain the working of Stair case A/D converter.

SECTION – C (30X10=30)

Answer any **THREE** Questions

16. Describe the structure, principle of operation, drain characteristics of N-channel JFET.

17. Describe the procedure for solving a second order differential equation using analog circuit. How are the initial conditions set up?

18. Explain the construction, working of Wien's Bridge oscillator using OP.AMP.

19. Write notes on a) Analog Phase detector b) Digital Phase detector.

20. Discuss with necessary block diagram, the working of successive approximation A/D converter.

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